



COURSE SLIDES

AI Agents for Trading (C177)

Course Code: C177

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Course Trainer

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Applied AI, quantitative research, and automation practitioner

DELIVERS

bounded AI agents, market-data validation, strategy research, backtesting, risk controls, and paper trading

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

- 1 Your name and one trading-research task that you would like an AI agent to help with.
- 2 One trading decision or action you would never allow an AI model to perform without human review.
- 3 One source of evidence you would require before trusting a trading claim.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

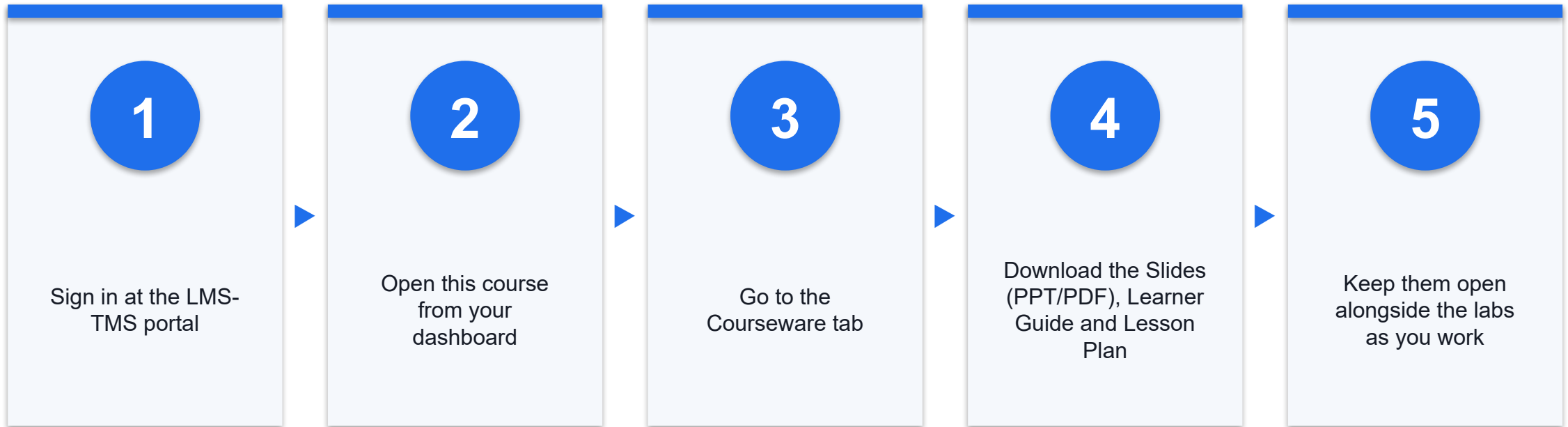
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Download Course Material



Portal: <https://lms-tms.tertiaryinfotech.com>

Lesson Plan — 1 Day, 8 Scheduled Hours

Course flow

- Day 1 — From Evidence-Led Research to a Controlled Paper Trade
 - Topic 1: From Trading Idea to Quality Market Data (Labs 1–2)
 - Topic 2: The Six-Step Trading Workflow and Staying in Control (Labs 3–4)

Timing & approach

- Daily timing
 - 9:00 am - 6:00 pm (1-hour lunch; two 15-minute tea breaks)
- Learning arc
 - Concept → guided build → verify → discuss → recap
- Connected practice
 - Four labs build one bounded Northstar SPY research pack from hypothesis to controlled paper evidence.

Learning Outcomes

1

Bounded Agent

Explain how a bounded AI trading agent combines a model, instructions, tools, state, evidence, guardrails, and human control without becoming an autonomous money manager.

2

Testable Hypothesis

Research a trading idea with an AI agent and convert it into a structured, falsifiable hypothesis with explicit rules, assumptions, evidence, and rejection criteria.

3

Quality Market Data

Connect the agent workflow to an authorised market-data source and validate coverage, timestamps, corporate-action adjustments, completeness, consistency, and freshness before use.

4

Backtest and Audit

Define deterministic trading rules, run an honest backtest, compare a benchmark, and audit data leakage, friction assumptions, overfitting, and out-of-sample behaviour.

5

Risk-Sized Ticket

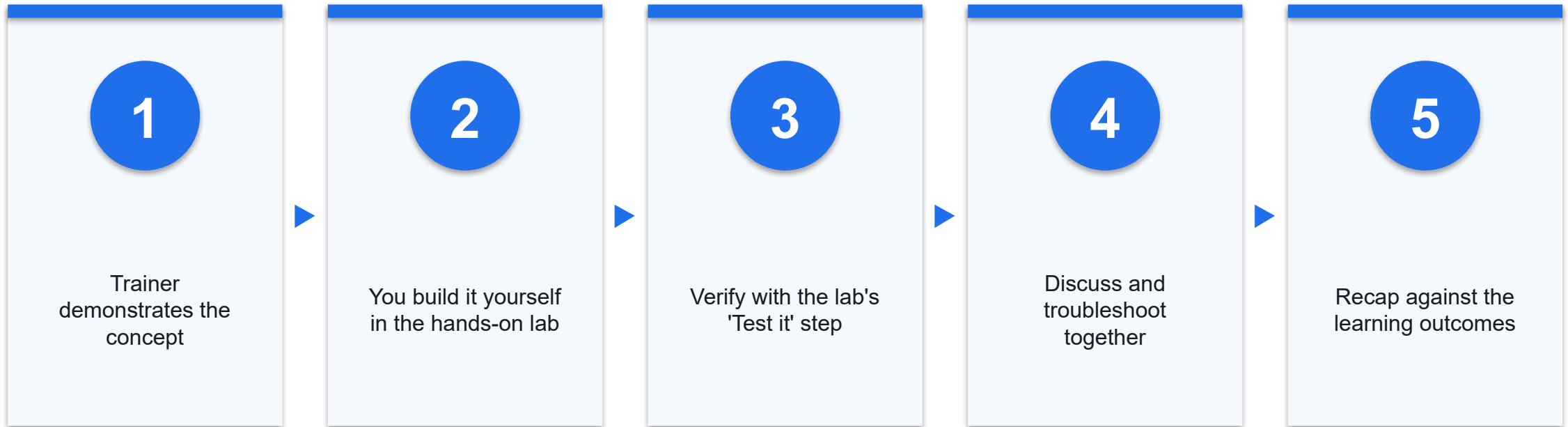
Calculate a bounded position size from account risk and stop distance, then prepare an auditable paper-order ticket with pre-trade limits and human approval evidence.

6

Controlled Paper Trade

Place and verify a paper-only trade through a controlled agent workflow while applying verify-then-trust checkpoints, logs, stop conditions, and rollback procedures.

How You'll Learn





CORE CONCEPTS

Course Fundamentals

The Evidence-Led Trading Agent

1

Research, not prediction

The agent turns an idea into testable evidence; it does not promise a market outcome.

2

Tools, not hidden arithmetic

Typed Python tools fetch, validate, backtest, size, and submit only within code-enforced bounds.

3

Paper, not live

Every order path is hard-wired to simulation and remains behind an explicit human gate.

4

Artifacts, not chat memory

Versioned JSON, CSV, Markdown, plots, fingerprints, and receipts form the audit trail.

The Six-Step Verify-Then-Trust Workflow

1

Define

Freeze the HYPER hypothesis and deterministic rules.

2

Validate

Accept only fit-for-purpose market data with recorded lineage.

3

Backtest

Use causal timing, visible frictions, a benchmark, and a holdout.

4

Audit

Check leakage, trials, stability, limitations, and decision criteria.

5

Size

Compute quantity from human-set risk and exposure limits.

6

Paper trade

Approve, submit to simulation, reconcile, verify, and roll back.

COURSE PRINCIPLE

A confident answer is not trading evidence.

Trust only the parts you can trace to a source, deterministic calculation, recorded decision, or verified paper receipt.

What You'll Build

Research Pack

- Structured HYPER hypothesis
- Cited research log
- Schema and version checks

Evidence Pack

- Market-data contract
- Quality report and fingerprint
- Backtest, trades, chart, and audit

Paper-Trade Pack

- Bounded risk calculation
- Dry-run ticket and approval
- Paper receipt and rollback record

How Every Lab Progresses

1 Start from a verified checkpoint.

2 Use the agent only within its evidence boundary.

3 Run deterministic checks before interpretation.

4 Save an observable artifact and fingerprint.

5 Stop, resolve, or continue through a named gate.

TOPIC 01

From Trading Idea to Quality Market Data

AI trading agents | research evidence | structured hypotheses | market-data connections | data-quality gates

Key Concepts — From Trading Idea to Quality Market Data

1

Bounded agency

The model may research and recommend, but deterministic tools calculate, controls constrain, and a person owns consequential decisions.

2

Falsifiable hypothesis

A useful idea states the asset, data, signal, entry, exit, horizon, benchmark, costs, risks, and rejection rule before testing.

3

Evidence chain

Every material claim links to a source, timestamp, tool result, assumption, or named owner instead of model confidence.

4

Market-data contract

The workflow records provider, feed, symbol, timezone, session, adjustment, interval, coverage, and intended use.

5

Quality gates

Schema, chronology, uniqueness, completeness, OHLC consistency, volume, freshness, and corporate actions are checked before analysis.

6

Human control

The agent stops when evidence is missing, conflicts remain, data quality is weak, or an action exceeds paper-only authority.

Chatbot, Workflow, Agent, or Autopilot?

Bounded research system

- Chat explains; workflow follows fixed rules
- Agent chooses among narrow read-only tools
- Code calculates and validates deterministically
- A person approves any paper order

Unsafe autopilot claim

- Model invents or changes trading rules mid-run
- One fluent answer is treated as evidence
- Keys or live-order endpoints are exposed
- No stop condition, audit trail, or owner exists

Seven Building Blocks of a Bounded Trading Agent

1

Goal

Turn one idea into a traceable research pack, not a buy or sell promise.

2

Model

Interprets questions, plans tool use, and drafts explanations; it is not the numeric authority.

3

Instructions

Define evidence, output schema, paper-only boundary, stop rules, and escalation.

4

Tools

Read market data, validate it, run a backtest, size risk, and prepare or submit a paper ticket.

Seven Building Blocks of a Bounded Trading Agent (cont.)

5

State

Versioned JSON, CSV, and logs carry verified facts between labs.

6

Guardrails

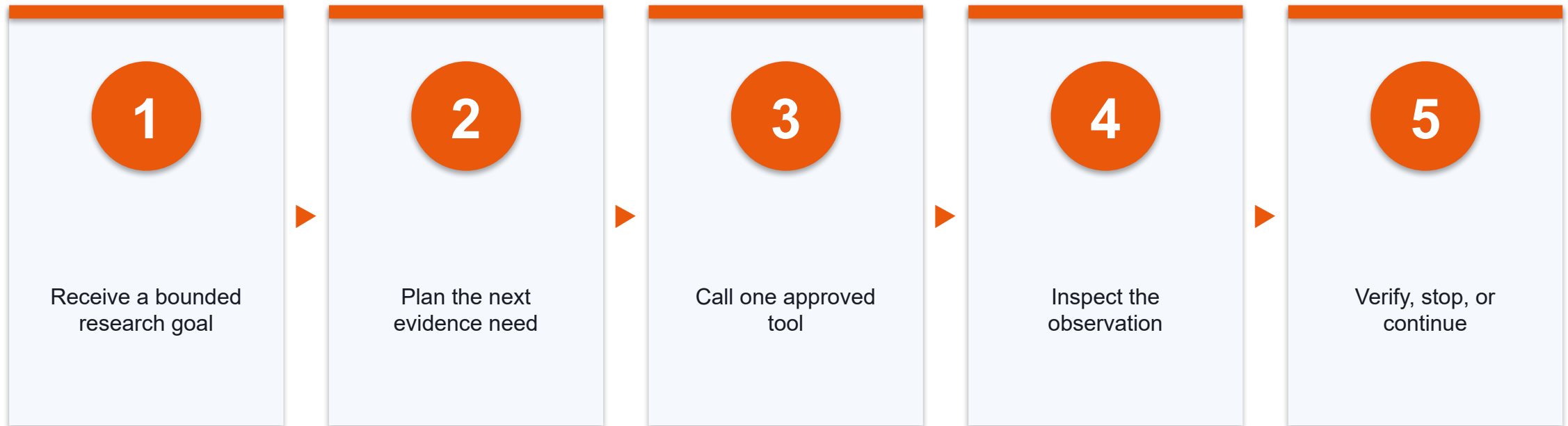
Allowlisted symbols, bounded dates, quantity caps, paper endpoint, dry run, and approval phrase.

7

Human owner

Reviews exceptions, approves the paper ticket, and decides whether research continues.

The Research Agent Loop



A Trading Idea Is Not Yet a Hypothesis

Vague idea

- Buy when momentum looks strong
- Use a good stock and recent data
- Exit when conditions change
- The strategy should beat the market

Structured hypothesis

- SPY daily adjusted bars, fixed date range
- 20-day SMA crosses above 50-day SMA after close
- Long next session; exit on opposite cross
- Compare with buy-and-hold after stated frictions

The HYPER Hypothesis Contract

1

H - Hypothesis

One falsifiable sentence and the market rationale it is intended to test.

2

Y - Yardstick

Benchmark, metrics, minimum sample, and rejection criteria defined in advance.

3

P - Parameters

Asset, interval, lookbacks, entry, exit, position state, and date range.

4

E - Evidence

Source URLs, market-data contract, assumptions, and OWNER_TO_VERIFY items.

5

R - Risks

Look-ahead, overfitting, costs, liquidity, regime change, model error, and operational failure.

Research Evidence: Stronger and Weaker Uses

Stronger evidence practice

- Exchange, regulator, provider, or official SDK documentation
- Direct dataset metadata and timestamped tool output
- Independent source used to cross-check a material claim
- Citation, access date, scope, and limitation recorded

Weaker evidence practice

- Uncited model memory or confident prose
- Screenshot with no source, time, or feed
- Promotional performance claim treated as proof
- Later edits that overwrite the original reasoning

An Agent Output Needs a Contract

1

Claim

What is being proposed, in one falsifiable sentence.

2

Known evidence

Source-backed facts and exact tool observations.

3

Assumptions

Choices that shape the result but are not measured facts.

4

Unknowns

Missing or conflicting items labelled OWNER_TO_VERIFY.

5

Next tool

The single approved action needed to reduce uncertainty.

6

Stop reason

Why the run cannot safely continue, when applicable.

What a Daily OHLCV Bar Represents

1

Timestamp

The labelled interval and timezone; it must match the intended trading session.

2

Open

First eligible trade price represented by the bar's aggregation rules.

3

High

Highest represented trade price; it should not be below open or close.

4

Low

Lowest represented trade price; it should not be above open or close.

5

Close

Final represented trade price; adjusted and raw closes answer different questions.

6

Volume

Aggregated eligible share volume; feed coverage affects the observed value.

Feed Coverage Changes What You Observe

Single-exchange feed

- Lower-cost access and useful for initial app testing
- Only trades and quotes from the named venue
- Volume and intraday prices may differ from the wider market
- Scope must be recorded in the data contract

Consolidated feed

- Combines eligible activity across US exchanges
- Broader price, quote, and volume representation
- May require a paid entitlement or provider plan
- Still requires timezone, session, and adjustment checks

Time, Session, and Signal Availability

Known at decision time

- Completed bars with unambiguous timezone
- Signal computed after the bar closes
- Order scheduled for the next eligible session
- Calendar and session assumptions recorded

Known only later

- Current bar's final close before it occurs
- Next session open used to decide today's signal
- Revised data silently substituted after the run
- Mixed local, exchange, and UTC dates

Corporate Actions and Adjusted Prices

1

Split

Share count and quoted price change mechanically; an unadjusted series can show a false crash.

2

Dividend

Cash distribution affects total return even when the price series alone does not show it.

3

Adjustment mode

Raw, split-adjusted, or fully adjusted data must be named in the contract.

4

Benchmark consistency

Strategy and benchmark must use comparable return and adjustment assumptions.

5

Point-in-time concern

Current symbol mappings and revised histories can leak later knowledge into old dates.

6

Verification

Inspect provider metadata and a known event before accepting the series.

The Market-Data Contract

1

Provider and endpoint

Who supplied the data and which documented route returned it.

2

Feed and entitlement

IEX, SIP, delayed, or another feed plus the account's access scope.

3

Instrument

Symbol, asset type, venue context, and currency.

4

Time definition

Interval, start, end, timezone, session, and as-of date.

5

Transformations

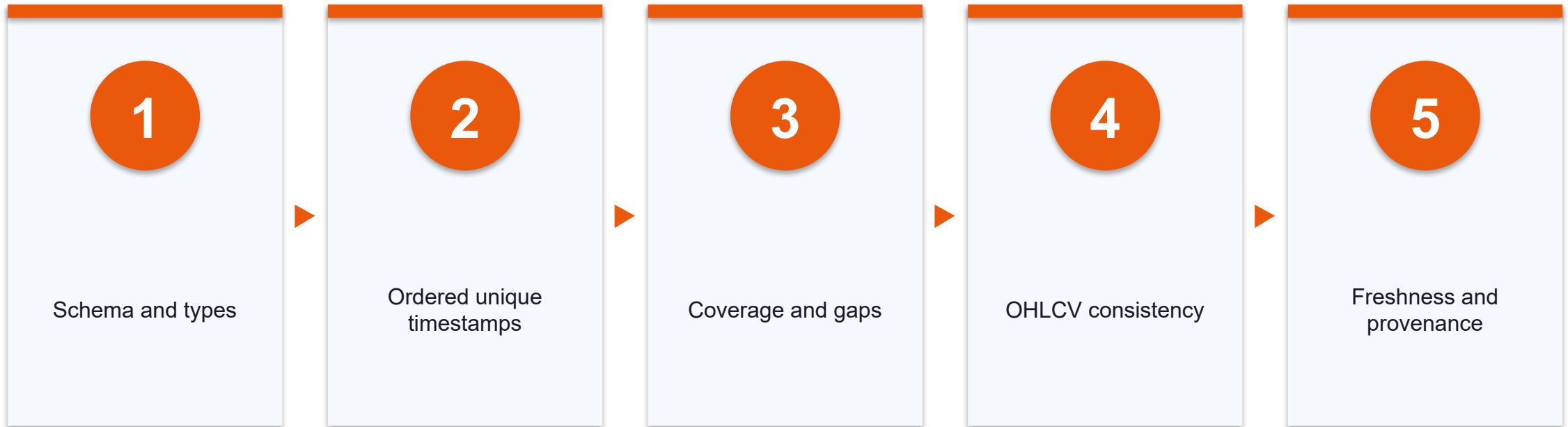
Adjustment, filtering, resampling, missing-value handling, and version.

6

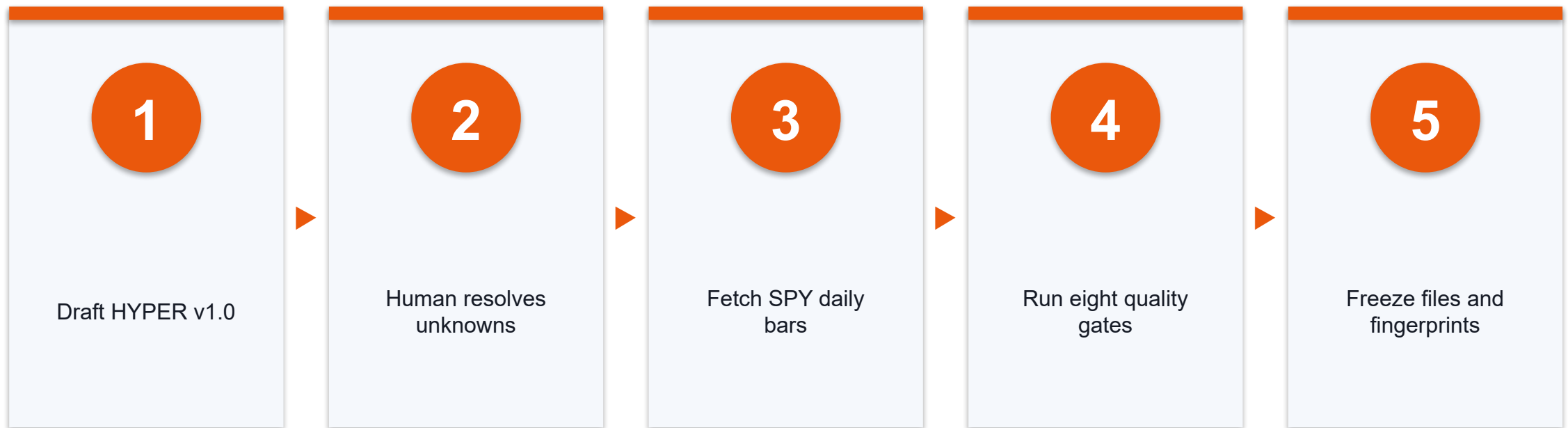
Intended use

Educational daily-bar research, not a live executable-price guarantee.

Eight Data-Quality Gates



Worked Example: Freeze the SPY Research Pack



Hands-On Labs — From Trading Idea to Quality Market Data

Lab 1

- Research an Idea and Freeze a Testable Hypothesis

Lab 2

- Fetch Market Data and Prove It Is Fit for Purpose

Research an Idea and Freeze a Testable Hypothesis

LAB 1

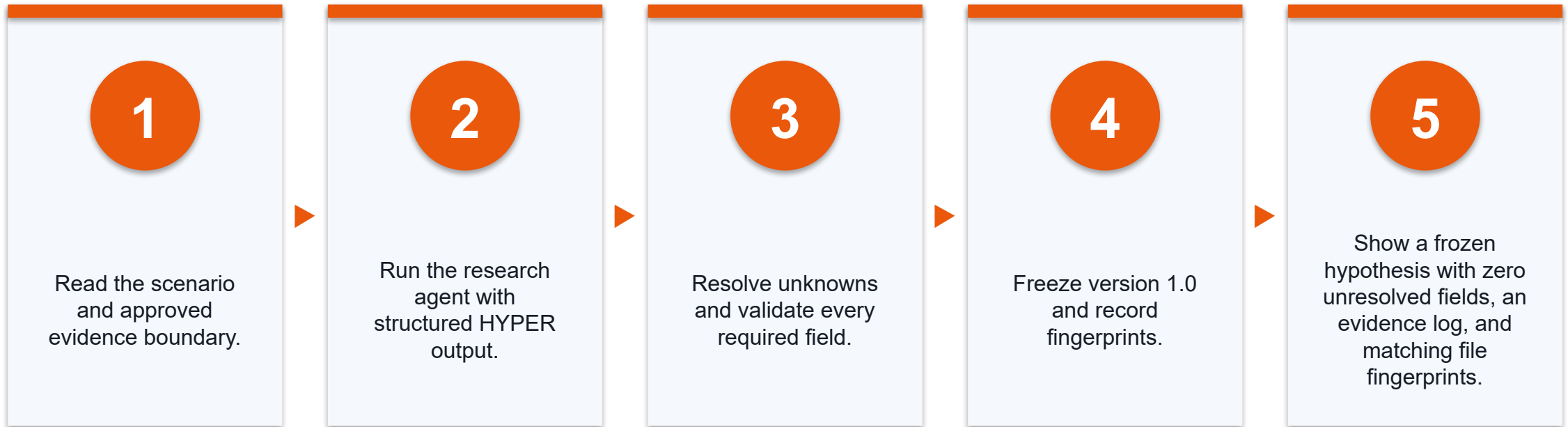
Start the connected SPY research scenario by running an OpenAI Agents SDK research agent with read-only course-source tools and structured output. Separate evidence, assumptions, and unknowns; resolve the required human-owned fields; validate the schema; then freeze version 1.0 with fingerprints before any market data is fetched.

You'll build

C177-trading-agent-pack/01-hypothesis/ containing **hypothesis_spec.json**, **research_log.md**, **agent_run_manifest.json**, **validation_report.json**, and **fingerprints.json** for the frozen SPY 20/50-day moving-average hypothesis

Resources: Use the matching lab's prerequisites and supplied files.

Lab 1 — Key Route



Fetch Market Data and Prove It Is Fit for Purpose

LAB 2

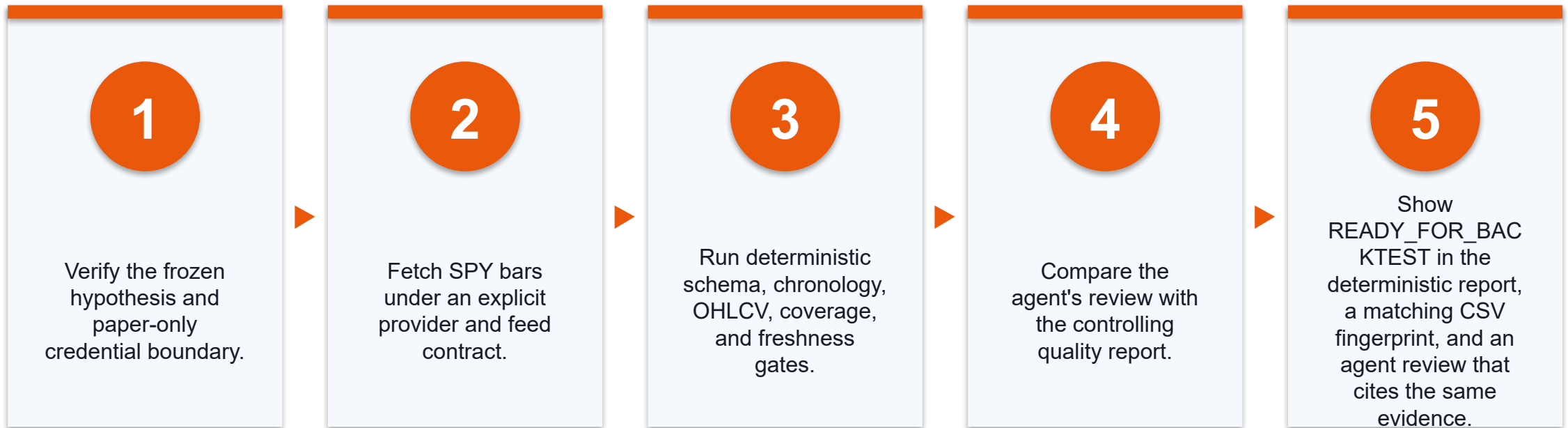
Verify the frozen hypothesis, connect to Alpaca's historical stock-data client, fetch the declared SPY daily bars, and produce a source contract, CSV, quality report, and fingerprint. Then give the research agent read-only access to the structured quality report so it can recommend READY_FOR_BACKTEST or HOLD_DATA without changing any numeric observation.

You'll build

C177-trading-agent-pack/02-market-data/ containing `spy_daily.csv`, `market_data_contract.json`, `data_quality_report.json`, `data_agent_review.json`, `retrieval_manifest.json`, and `fingerprints.json`

Resources: Use the matching lab's prerequisites and supplied files.

Lab 2 — Key Route



Recap — From Trading Idea to Quality Market Data

1

You can now: LO1 and LO2: operate a bounded tool-using research agent and convert one trading idea into a cited, structured, falsifiable HYPER hypothesis

2

You can now: LO3: connect the workflow to authorised Alpaca market data, record the data contract, run deterministic quality gates, and let the agent interpret only the verified report

TOPIC 02

The Six-Step Trading Workflow and Staying in Control

rule specification | honest backtesting | audit checks | risk sizing | paper orders | verify-then-trust checkpoints

KEY CONCEPTS

Key Concepts — The Six-Step Trading Workflow and Staying in Control

1

Rules before results

Entry, exit, timing, sizing, costs, benchmark, and rejection criteria are frozen before the engine runs.

2

Deterministic core

Code owns calculations and enforcement; the model interprets, organises evidence, and explains uncertainty.

3

Backtest realism

Signals are shifted, data is split, and fees, spread, slippage, fills, and benchmark assumptions are visible.

4

Risk budget

Quantity follows a capped account-risk budget and stop distance, never a model's confidence score.

5

Paper-only execution

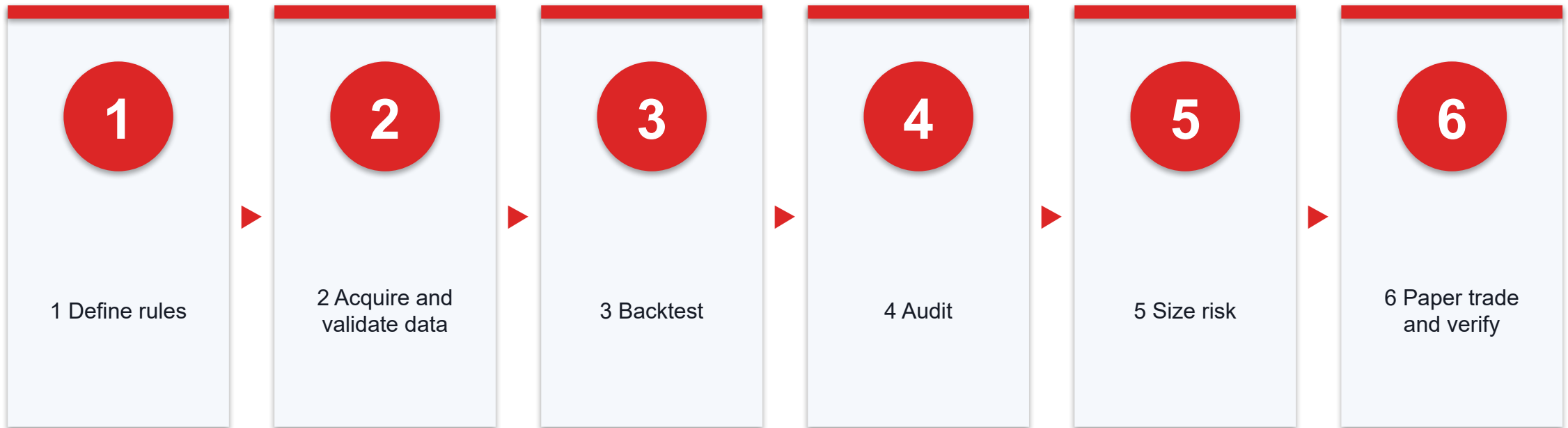
The supplied client uses the paper endpoint, a symbol allowlist, dry run, quantity cap, and explicit approval token.

6

Verify then trust

Every stage produces checks, logs, fingerprints, a decision, and a safe rejoin point.

The Six-Step Controlled Trading Workflow



Rule Specification: Remove Every Hidden Choice

1

Universe

One named asset or a point-in-time selection rule.

2

Signal

Exact formula, lookback, bar interval, and data field.

3

Timing

When the signal becomes known and when the order is assumed to execute.

4

Entry and exit

Conditions, position state, and conflict resolution.

Rule Specification: Remove Every Hidden Choice (cont.)

5

Sizing

Fixed rule independent of model enthusiasm.

6

Friction

Fees, spread or slippage, and fill assumptions.

7

Benchmark

A comparable alternative such as buy-and-hold.

8

Decision

Predeclared evidence for CONTINUE, HOLD, or REVISE.

Look-Ahead Bias: The One-Bar Test

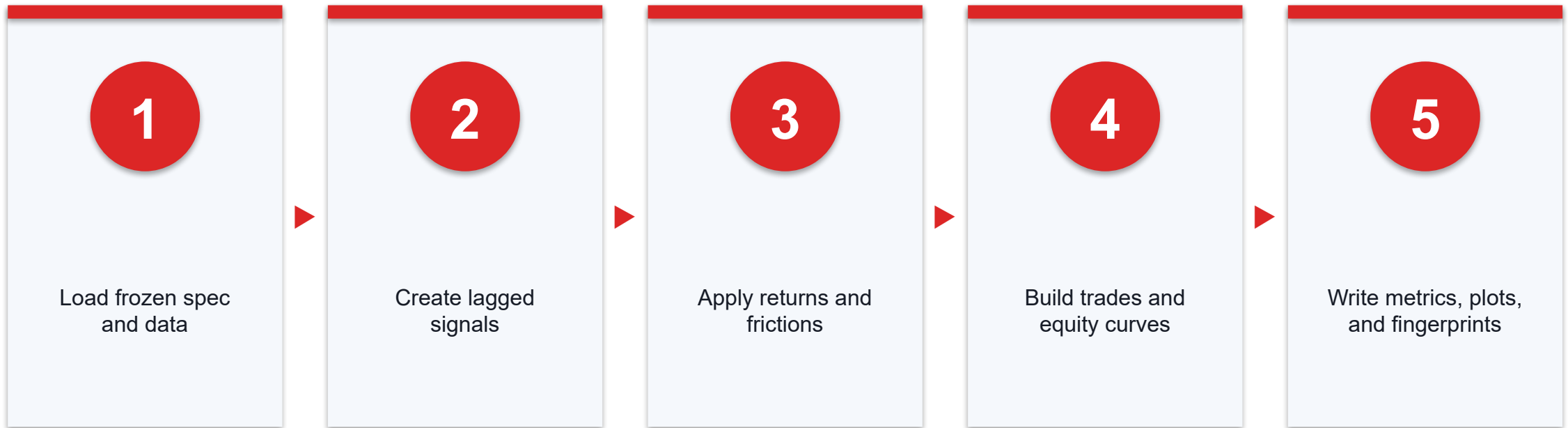
Biased implementation

- Today's close creates today's signal
- Today's signal receives today's close-to-close return
- Final high or low is used before the bar completes
- Future revisions influence an earlier decision

Causal implementation

- Completed bar creates the signal after close
- Signal is shifted before the next return is applied
- Execution timing is written in the rule contract
- Inputs and code versions are fingerprinted

Backtest Engine: Inputs to Evidence



Development Window and Out-of-Sample Window

Development segment

- Used to implement and debug the fixed rule
- Can reveal obvious data or code errors
- Results may influence design choices
- Must not be presented as untouched evidence

Out-of-sample segment

- Held back until the specification is frozen
- Used once for a cleaner generalisation check
- Compared with benchmark and decision criteria
- Poor evidence leads to HOLD, not repeated tuning

Read Metrics as a System, Not a Score

1

Total return

Growth over the window; it depends strongly on start and end dates.

2

Annualised return

Compounded rate normalised for time; unstable with short samples.

3

Volatility

Dispersion of returns, not a complete measure of loss risk.

4

Sharpe ratio

Excess return per unit of volatility under stated assumptions.

Read Metrics as a System, Not a Score (cont.)

5

Maximum drawdown

Largest peak-to-trough decline in the simulated equity curve.

6

Turnover

How often exposure changes; it drives friction sensitivity.

7

Trade count

A tiny sample cannot support a stable hit-rate story.

8

Benchmark gap

Strategy result minus a comparable passive alternative.

Frictionless Results and More Realistic Results

Frictionless simplification

- Every order fills immediately at the chosen bar price
- No spread, fee, slippage, latency, or market impact
- Unlimited liquidity at any quantity
- Useful only as an upper-bound diagnostic

Declared reality assumptions

- Turnover incurs stated fee and slippage basis points
- Execution timing and price field are explicit
- Paper limitations and liquidity are disclosed
- Sensitivity cases show how conclusions change

Overfitting Red Flags

1

Many trials

Dozens of unreported variants make the best result look rarer than it is.

2

Tiny sample

Few trades or one market regime cannot support a stable conclusion.

3

Sharp optimum

Only one exact parameter works while nearby values collapse.

4

No benchmark

A complex strategy may merely reproduce market exposure.

5

Hidden exclusions

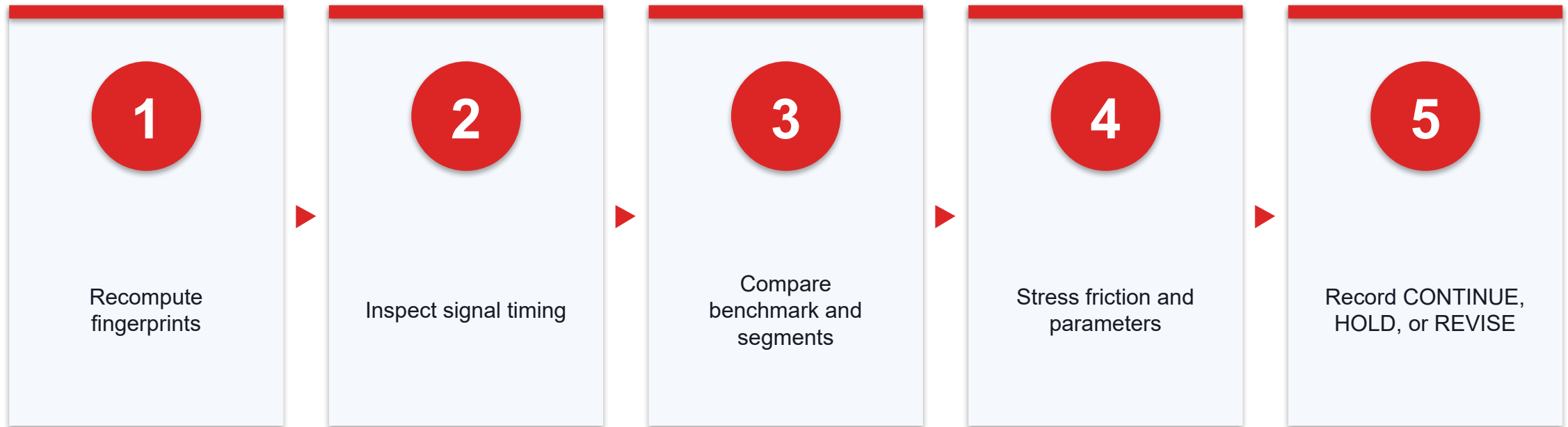
Changing assets or dates after inspection creates selection bias.

6

Holdout erosion

Repeatedly tuning on the same out-of-sample period destroys its role.

The Backtest Audit Trail



Model Work and Deterministic Work

AI model may

- Summarise sources and organise a hypothesis draft
- Choose the next approved read-only tool
- Explain metrics and surface uncertainty
- Draft a HOLD or CONTINUE rationale for review

Code or human must

- Validate schema, timestamps, prices, and fingerprints
- Compute signals, returns, costs, and position size
- Enforce paper endpoint, allowlist, caps, and dry run
- Approve the ticket and own the decision

Position Size Comes from Risk Budget and Stop Distance

1

Account equity

The paper account value used only for the exercise.

2

Risk fraction

A small predeclared fraction chosen by the human owner, not the agent.

3

Risk budget

Account equity multiplied by risk fraction.

4

Per-share risk

Absolute difference between planned entry and protective stop.

5

Risk quantity

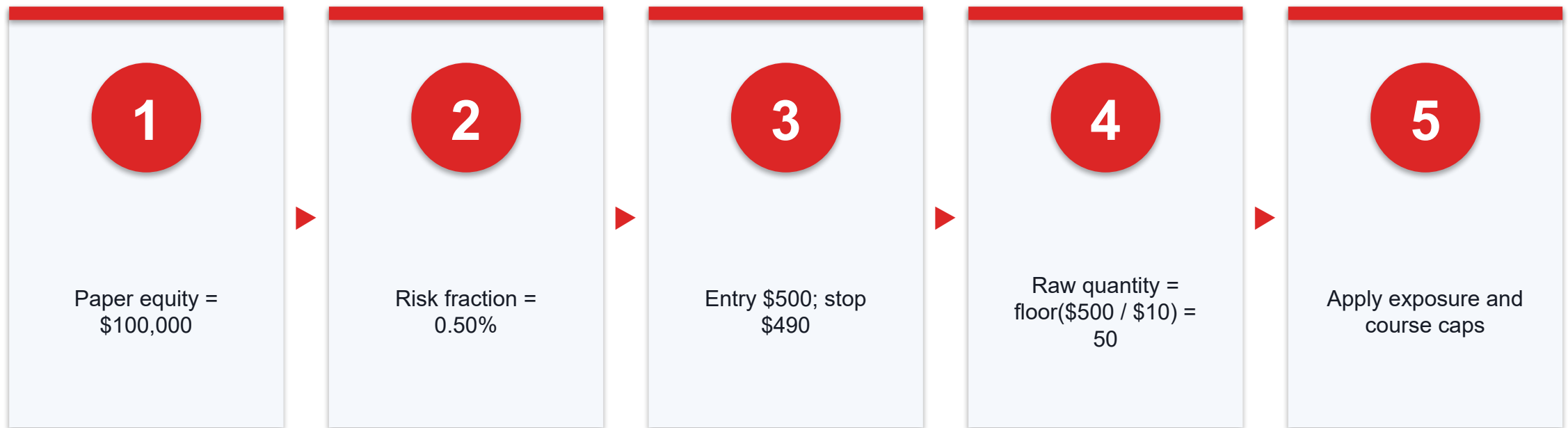
Floor of risk budget divided by per-share risk.

6

Exposure cap

A second ceiling based on maximum position value and available buying power.

Worked Position-Size Example



The Pre-Trade Paper Ticket

1

Instrument

Allowlisted symbol, asset type, side, and paper environment.

2

Evidence

Accepted hypothesis, data, backtest, audit decision, and fingerprints.

3

Quantity

Risk calculation, exposure cap, buying-power check, and hard maximum.

4

Order

Type, time in force, reference price, and paper endpoint.

5

Limits

No live route, no secret in logs, no unsupported symbol, no blank approval.

6

Owner

Reviewer, approval time, approval token, and reason for the exercise.

Paper Trading Is Useful and Incomplete

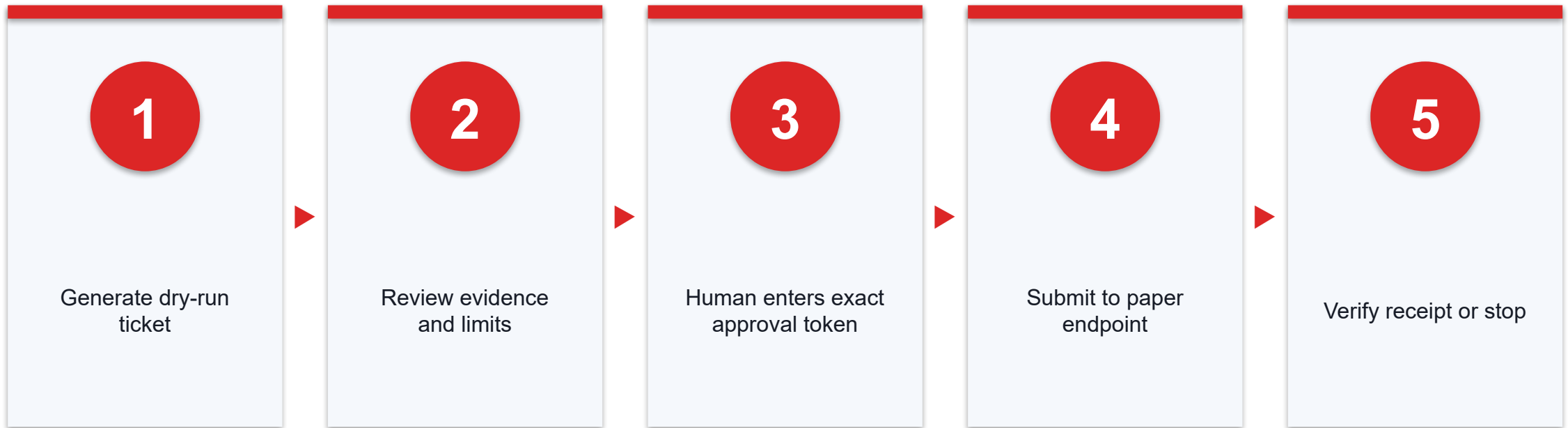
What paper trading exercises

- Authentication, request schema, account checks, and order lifecycle
- Real-time data handling and software error paths
- Logging, approval, idempotency, and receipt verification
- Operational rehearsal without risking capital

What it may not reproduce

- Market impact, information leakage, or latency slippage
- Queue position and realistic available liquidity
- All fees, dividends, price improvement, or live behaviour
- Future profitability or suitability for real money

Human Gate and Safe Submission



Verify-Then-Trust Checkpoints

1

Hypothesis gate

Schema valid, assumptions separated, unknowns owned, version frozen.

2

Data gate

Contract complete, quality checks recorded, fingerprint accepted.

3

Backtest gate

Timing causal, benchmark present, segments and frictions reported.

4

Audit gate

Leakage, trials, stability, limitations, and decision documented.

5

Risk gate

Quantity recomputed from approved inputs and bounded by caps.

6

Execution gate

Paper endpoint, human approval, unique ID, receipt, and reconciliation.

Failure, Pause, Reconcile, and Roll Back

1

Tool failure

Record the error; do not fabricate a result or continue from stale state.

2

Data drift

Freeze the new dataset separately and rerun quality and backtest evidence.

3

Duplicate risk

Query the client order ID before retrying a timed-out request.

4

Unexpected fill

Stop the workflow, inspect the paper account, and preserve the receipt.

5

Model change

Rerun stable prompts and tool-choice checks before accepting explanations.

6

Rollback

Cancel open paper orders, disable submission, and restore the last verified version.

Hands-On Labs — The Six-Step Trading Workflow and Staying in Control

Lab 3

- Backtest the Rules and Audit the Evidence

Lab 4

- Size Risk, Approve, and Verify a Paper Trade

Backtest the Rules and Audit the Evidence

LAB 3

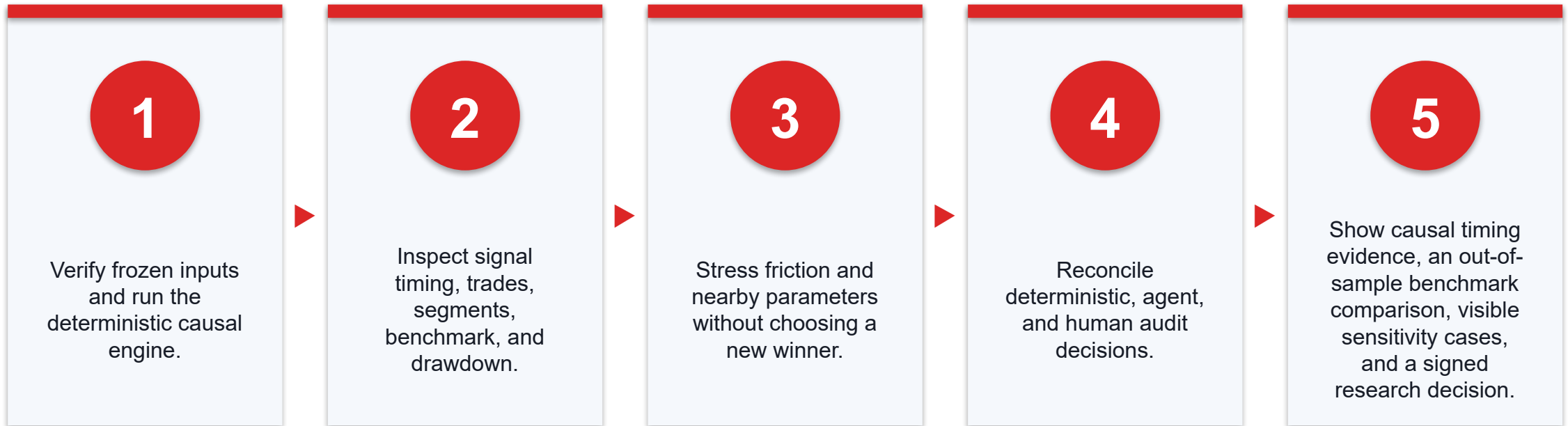
Verify the accepted hypothesis and market-data fingerprints, run the fixed 20/50-day crossover engine with a one-bar signal shift and declared frictions, then inspect trades, development and out-of-sample metrics, benchmark results, drawdown, and sensitivity checks. The agent receives only read-only report tools and drafts a review that the human auditor must reconcile with deterministic checks.

You'll build

C177-trading-agent-pack/03-backtest-audit/ containing **strategy_spec.json**, **signal_sample.csv**, **trades.csv**, **equity_curve.csv**, **equity_curve.png**, **backtest_metrics.json**, **sensitivity_report.json**, **audit_log.md**, **backtest_agent_review.json**, **run_manifest.json**, and **fingerprints.json**

Resources: Use the matching lab's prerequisites and supplied files.

Lab 3 — Key Route



Size Risk, Approve, and Verify a Paper Trade

LAB 4

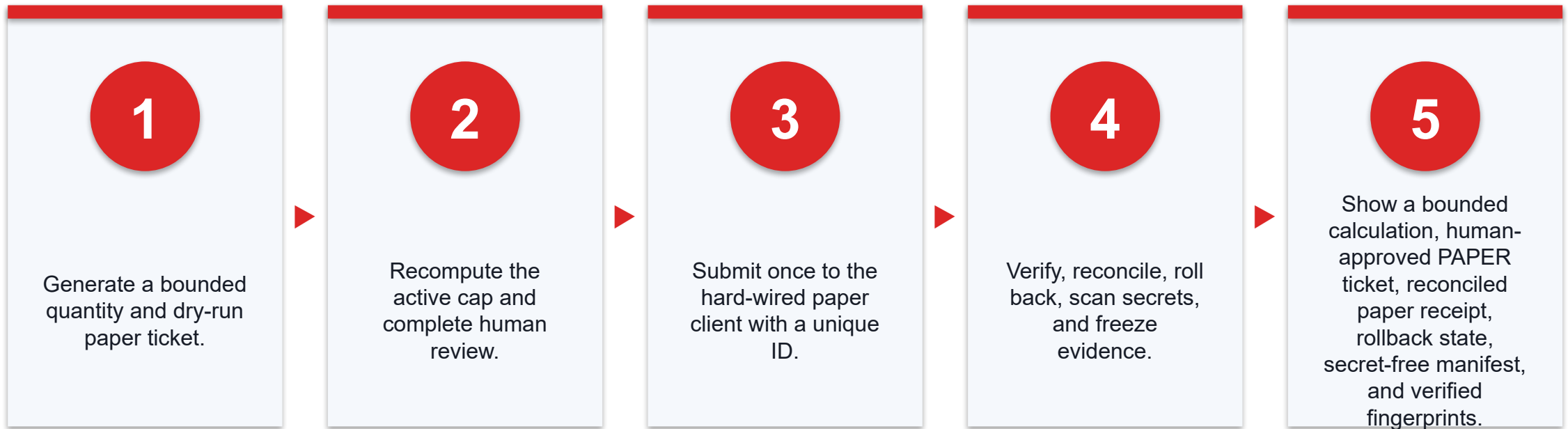
Use the accepted research evidence to calculate quantity from a human-set paper risk budget, stop distance, exposure ceiling, buying power, and hard course cap. Generate a dry-run ticket, let the agent perform a read-only evidence review, obtain explicit human approval, submit through an Alpaca TradingClient constructed with paper=True, reconcile the receipt, and close with a secret-free final manifest.

You'll build

C177-trading-agent-pack/04-paper-trade/ containing **risk_calculation.json**, **paper_order_ticket.json**, **ticket_agent_review.json**, **human_approval.md**, **submission_attempt.json**, a paper receipt or structured error, **reconciliation_log.md**, **rollback_record.md**, **final_manifest.json**, and **fingerprints.json**

Resources: Use the matching lab's prerequisites and supplied files.

Lab 4 — Key Route



Recap — The Six-Step Trading Workflow and Staying in Control

1

You can now: LO4: run a causal deterministic backtest, compare a benchmark and holdout, stress stated assumptions, and record an auditable research decision

2

You can now: LO5 and LO6: calculate a bounded quantity, generate and review a paper-order ticket, apply the human gate, submit only to simulation, and verify or roll back the result



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Bounded Agent

Explain how a bounded AI trading agent combines a model, instructions, tools, state, evidence, guardrails, and human control without becoming an autonomous money manager.

2

Testable Hypothesis

Research a trading idea with an AI agent and convert it into a structured, falsifiable hypothesis with explicit rules, assumptions, evidence, and rejection criteria.

3

Quality Market Data

Connect the agent workflow to an authorised market-data source and validate coverage, timestamps, corporate-action adjustments, completeness, consistency, and freshness before use.

4

Backtest and Audit

Define deterministic trading rules, run an honest backtest, compare a benchmark, and audit data leakage, friction assumptions, overfitting, and out-of-sample behaviour.

5

Risk-Sized Ticket

Calculate a bounded position size from account risk and stop distance, then prepare an auditable paper-order ticket with pre-trade limits and human approval evidence.

6

Controlled Paper Trade

Place and verify a paper-only trade through a controlled agent workflow while applying verify-then-trust checkpoints, logs, stop conditions, and rollback procedures.

Continue with Evidence, Not Automation Pressure

1 Reproduce the same run before changing the idea.

2 Version every rule, dataset, prompt, code, and decision change.

3 Increase realism and independent review before increasing autonomy.

4 Keep execution paper-only throughout the course and personal practice.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Your course material stays available at <https://lms-tms.tertiaryinfotech.com/>.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

KEEP VERIFYING BEFORE YOU TRUST

Thank You!

You can now direct a bounded AI agent through a traceable trading research cycle while keeping data, calculation, risk, and execution evidence under human control.